11. Quine and Strawson : Critique of Empiricism; Theory of Basic Particulars and Persons.
12. Cârvâka : Theory of Knowledge; Rejection of Transcendent Entities.
13. Jainism: Theory of Reality; Saptabhaôginaya; Bondage and Liberation.
14. Schools of Buddhism: Pratîtyasamutpâda; Ksanikavada, Nairâtmyavâda
15. Nyâya-Vaiúesika : Theory of Categories; Theory of Appearance; Theory of Pramâna; Self, Liberation; God; Proofs for the Existence of God; Theory of Causation; Atomistic Theory of Creation.
16. Sâmkhya: Prakrti; Purusa; Causation; Liberation
17. Yoga: Citta; Cittavrtti; Klesas; Samadhi; Kaivalya.
18. Mimâmsâ : Theory of Knowledge
19. Schools of Vedânta : Brahman; Îúvara; Âtman; Jiva; Jagat; Mâyâ; Avidyâ; Adhyâsa; Moksa; Aprthaksiddhi; Pancavidhabheda
20. Aurobindo: Evolution, Involution; Integral Yoga.

PAPER–II

Socio-Political Philosophy :

1. Social and Political Ideals : Equality, Justice, Liberty.
2. Sovereignty : Austin, Bodin, Laski, Kautilya.
3. Individual and State : Rights; Duties and Accountability
4. Forms of Government : Monarchy; Theocracy and Democracy.
5. Political Ideologies : Anarchism; Marxism and Socialism.
6. Humanism; Secularism; Multiculturalism.
7. Crime and Punishment: Corruption, Mass Violence, Genocide, Capital Punishment.
8. Development and Social Progress.
9. Gender Discrimination : Female Foeticide, Land and Property Rights; Empowernment.
10. Caste Discrimination: Gandhi and Ambedkar.

Philosophy of Religion :

1. Notions of God : Attributes; Relation to Man and the World. (Indian and Western).
2. Proofs for the Existence of God and their Critique (Indian and Western).
3. Problem of Evil.
4. Soul: Immortality; Rebirth and Liberation.
5. Reason, Revelation and Faith.
6. Religious Experience: Nature and Object (Indian and Western).
7. Religion without God.
8. Religion and Morality.
9. Religious Pluralism and the Problem of Absolute Truth.
10. Nature of Religious Language: Analogical and Symbolic; Cognitivist and Non- cognitive.

PHYSICS

PAPER–I

1. (a) Mechanics of Particles—Laws of motion; conservation of energy and momentum, applications to rotating frames, centripetal and Coriolis accelerations; Motion under a central force; Conservation of angular momentum, Kepler's laws; Fields and potentials; Gravitational field and potential due to spherical bodies, Gauss and Poisson equations, gravitational self-energy; Two-body problem; Reduced mass; Rutherford scattering; Centre of mass and laboratory reference frames.

(b) Mechanics of Rigid Bodies—System of particles; Centre of mass, angular momentum, equations of motion; Conservation theorems for energy, momentum and angular momentum; Elastic and inelastic collisions; Rigid body; Degrees of freedom, Euler's theorem, angular velocity, angular momentum, moments of inertia, theorems of parallel and perpendicular axes, equation of motion for rotation; Molecular rotations (as rigid bodies); Di and tri-atomic molecules; Precessional motion; top, gyroscope.

(c) Mechanics of Continuous Media—Elasticity, Hooke's law and elastic constants of isotropic solids and their inter-relation; Streamline (Laminar) flow, viscosity, Poiseuille's equation, Bernoulli's equation, Stokes' law and applications.

(d) Special Relativity—Michelson-Morley experiment and its implications; Lorentz transformations-length contraction, time dilation, addition of relativistic velocities, aberration and Doppler effect, mass-energy relation, simple applications to a decay process; Four dimensional momentum vector; Covariance of equations of physics.

2. Waves and Optics :

(a) Waves—Simple harmonic motion, damped oscillation, forced oscillation and resonance; Beats; Stationary waves in a string; Pulses and wave packets; Phase and group velocities; Reflection and Refraction from Huygens' principle.

(b) Geometrical Optics—Laws of reflection and refraction from Fermat's principle; Matrix method in paraxial optics-thin lens formula, nodal planes, system of two thin lenses, chromatic and spherical aberrations.

(c) Interference—Interference of light-Young's experiment, Newton's rings, interference by thin films, Michelson interferometer; Multiple beam interference and Fabry-Perot interferometer.

(d) Diffraction—Fraunhofer diffraction-single slit, double slit, diffraction grating, resolving power; Diffraction by a circular aperture and the Airy pattern; Fresnel diffraction: half-period zones and zone plates, circular aperture.